

Leopoldo Cuspinera

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SUMMARY

Data Scientist with 5+ years of experience in machine learning, graph analytics, and forecasting for power networks. PhD in Physics with expertise in Neo4j, Spark, and Databricks. Proven ability to lead projects that drive operational efficiency and data-driven decision-making. Seeking a leadership role combining hands-on data science with strategic impact.

EDUCATION

- **Durham University** Durham, England
PhD in Physics Oct 2015 – Jan 2020
- **Durham University** Durham, England
MSc in Strings Particles and Cosmology. Honours: Distinction Oct 2014 – Sep 2015
- **Benemerita Universidad Autonoma de Puebla** Puebla, Mexico
BSc. in Physics Aug 2009 – Jun 2014

EXPERIENCE

- **Camlin Group** Remote, Italy
Data Scientist Jun 2021 - Present
 - **Data Modelling and Neo4j Knowledge Graphs:** Led a team of subject matter experts and data engineers in designing data models and algorithms for ingesting, aggregating, cleansing, and validating time series measurements across the UK electrical grid. Built a Neo4j-based knowledge graph integrating over 15M nodes, 16M relationships, and 220M properties, enabling real-time tracking of time series from thousands of transformers and over 70k generators. Through direct customer engagement, data quality improved significantly, and key modeling approaches were adopted by the client, demonstrating the project's strategic impact.
 - **Neo4j and Graph Algorithms:** Developed a Python package to leverage Neo4j for analyzing physical network connectivity and power supply, integrating graph and ML algorithms to predict missing data, optimize network traversal, and track load distribution. Not only does this tool provides a graph analysis of the network, but also a map-based visibility, overlaying satellite and other geographic maps to visualize circuits and their underlying assets, enhancing spatial understanding for predictive maintenance and phase balancing optimization.
 - **Python, Spark, Databricks:** Graph and geospatial analysis to create profiles of transmission lines based on load consumption in UK's electrical grid at low voltages. Design and development of models that forecast electrical faults through time series analysis, signal processing and different ML algorithms. Provision of advice to Distribution Network Operators in prioritising correctly the required maintenance to reduce the economical impact faulty services may cause.
 - **Tableau:** Creation of Stories and Dashboards that summarise convoluted information about the historical state of the network thus giving visibility of the network to stakeholders and providing a starting point for quick investigations.
- **Esosphaera S.R.L.** Treviso, Italy
Data Scientist Sep 2020 - Jun 2021
 - **Python:** Created ETL pipelines that queried different tables of the main database via Postgres, created features with pandas, updated the curated data on S3 buckets and made use of Athena to feed Tableau dashboards.
 - **Tableau:** Created several interactive Dashboards that allowed both salesmen and stakeholders in understanding the evolution of the most impactful interactions between end users and customers at a glance.
- **Perimeter Institute** Waterloo, Canada
Visiting Researcher Jan 2016 - Mar 2019
 - **Mathematica, Python:** Made extensive use of both Mathematica and Python to solve the Partial Differential Equations that describe vacuum decays around higher dimensional black holes. Worked in an international collaboration that had as a result two of the papers I published during my PhD.

- **Guest speaker:** Gave a seminar explaining the calculations we made to obtain the results of our work.
- **Durham University** Durham, England
Teaching Assistant *Oct 2015 - Nov 2019*
 - **Maths and stats lab:** guided students with different academic backgrounds in finding their own answers to mathematical and statistical problems. Due to the positive impact our team had on the students, we obtained the 'Student employee of the year (2019)' award.
 - **Tutor:** Helped students in understanding concepts in Quantum Mechanics, Linear Algebra, Statistics and Calculus
- **Deutsches Elektronen-Synchrotron** Hamburg, Germany
Internship *Jun 2014 - Sep 2014*
 - **C++:** Studied the predictions of different Monte Carlo simulators on diffractive dissociations of proton-proton collisions and compared with real data measured at LHC. This was my first time being exposed to C++, which demonstrates I can quickly pick up a programming language, develop a project and provide results in short time.

SKILLS

- **Technical Skills**
 - **Programming and Data Tools:** Python, SQL, Git, Neo4j, Spark, Pandas, Scikit-learn, TensorFlow, Keras, Tableau, Bokeh, Seaborn, Mathematica, LaTeX, AWS (boto3), Bash, C++, Excel.
 - **Mathematical and Analytical Expertise:** Machine Learning (SVM, KNN, Random Forests, NN, CNN, ARIMA), Graph Algorithms, Time Series Analysis, Linear Algebra, Pattern Identification, Quantitative Analysis.
- **Soft Skills and Research:** Project Ownership, Stakeholder Communication, Strategy, Decision-Making, Team Leadership, Cross-functional Collaboration, Public Speaking, Communication, Self-Training, Research.
- **Languages:** English, Spanish, Italian.

PUBLICATIONS

- **Black holes, vacuum decay and thermodynamics** *Jan 2020*
PhD thesis
- **Are Superentropic black holes superentropic?** *Nov 2019*
Journal of High Energy Physics
- **Higgs vacuum decay in a braneworld** *Nov 2019*
International Journal of Modern Physics D
- **Higgs Vacuum Decay from Particle collisions?** *Jan 2019*
Physical Review D